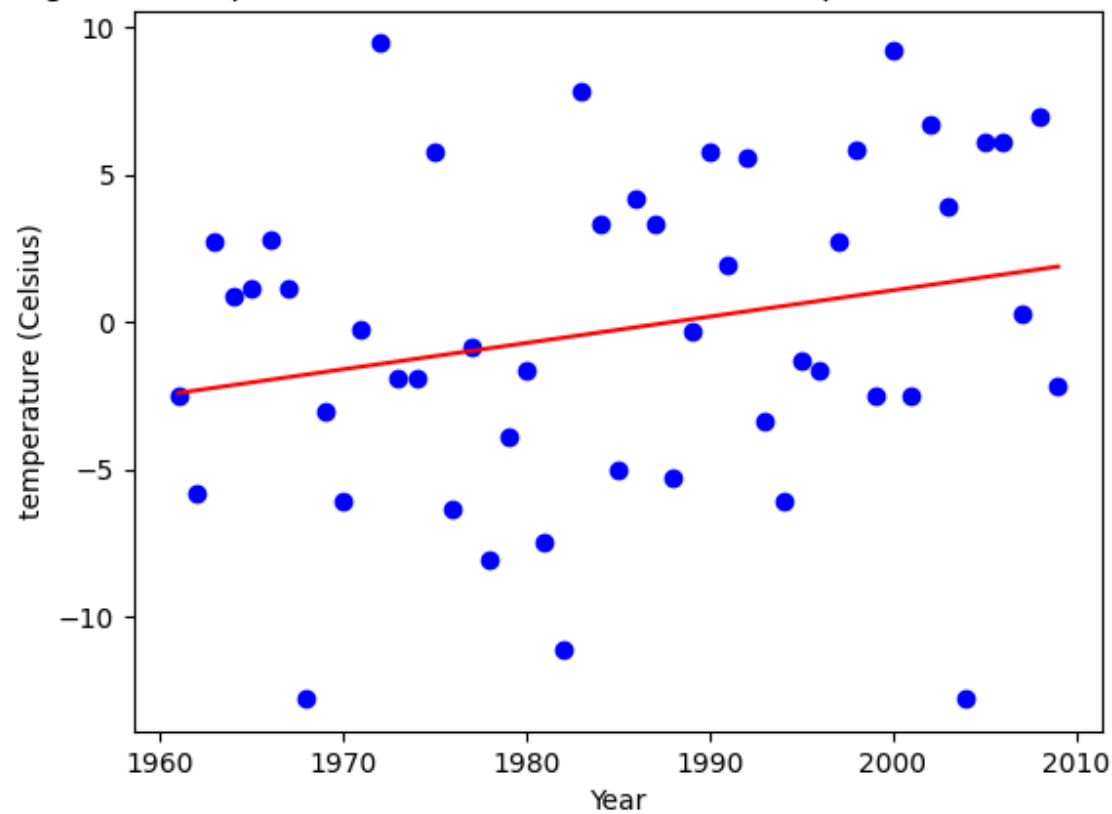
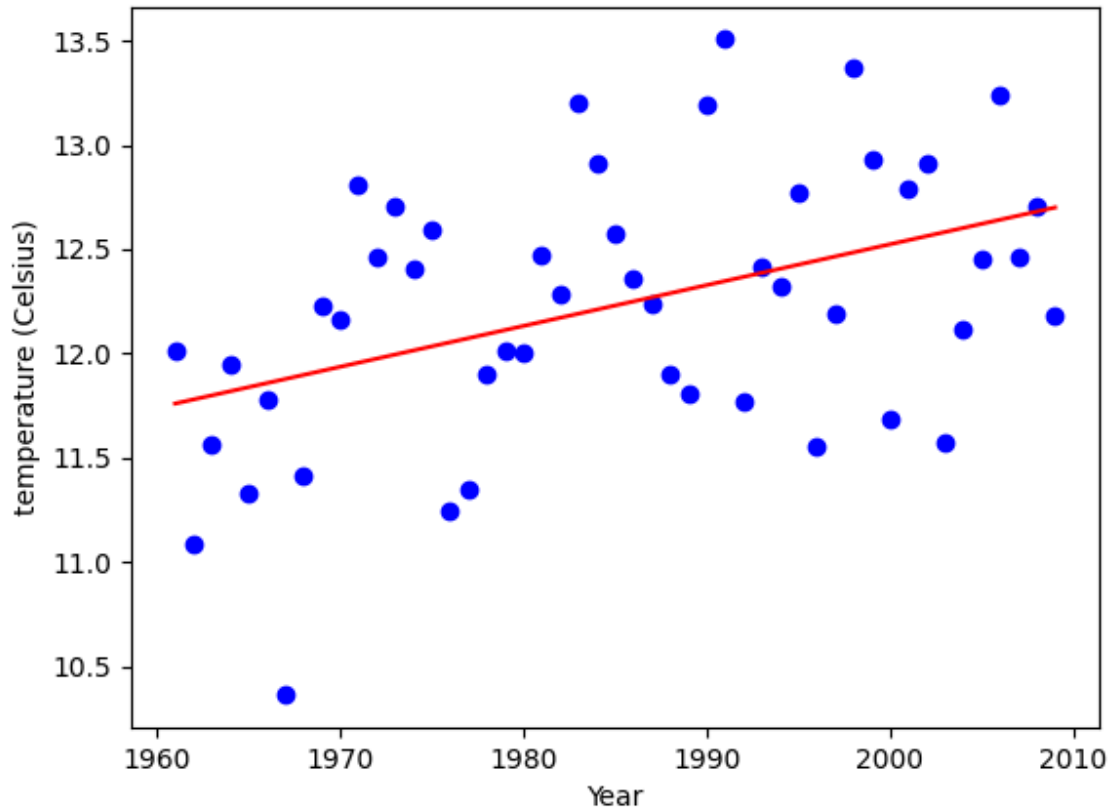


PART A:

degree 1, R-square 0.05347527266200136, SE/slope 0.6136779927980907



degree 1, R-square 0.18895346977477967, SE/slope 0.3022014339205546!



- What difference does choosing a specific day to plot the data for versus calculating the yearly average have on our graphs (i.e., in terms of the R^2 values and the fit of the resulting curves)? Interpret the results.

Using average yearly temperature appears to produce a better fit than using the temperature on January 10th as the y-value. The R^2 of plot two is higher than plot one, proving a better fit. It appears that averaging out the temperatures minimizes random daily fluctuations and as a result produces a better model.

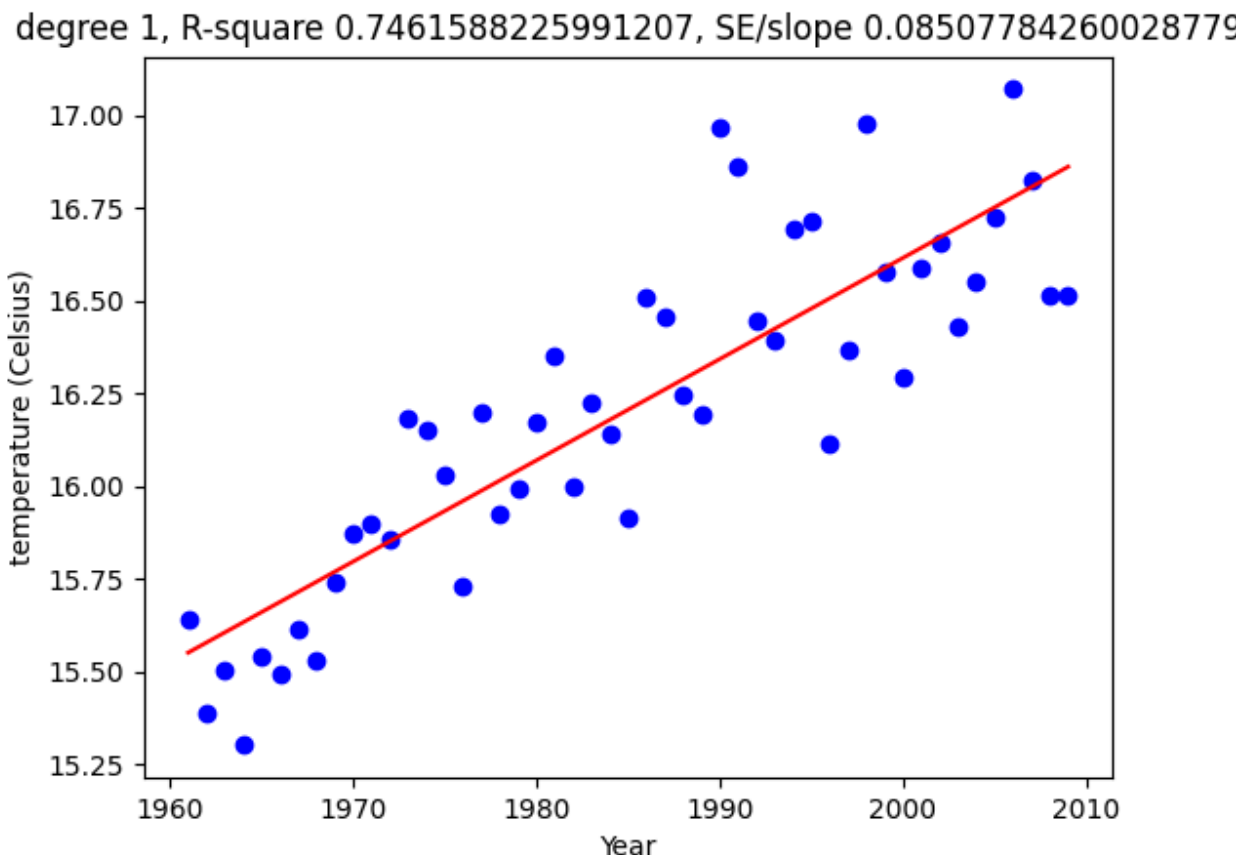
- Why do you think these graphs are so noisy? Which one is more noisy?

I believe the graphs are noisy because of the underlying nature of temperatures. Temperatures are naturally stochastic, hence the noise. The January 10th graph is more noisy than the average temperature graph as averaging yearly temperature serves to minimize random fluctuations. The temperature on January 10th doesn't represent the temperature throughout the year.

- How do these graphs support or contradict the claim that global warming is leading to an increase in temperature? The slope and the standard error-to-slope ratio could be helpful in thinking about this.

In both the average graph and January 10th graph there is a consistent increase in yearly temperature between 1961 and 2009 measured by the graphs different methods. The slope in both graphs is positive showing the temperature increase as the years increase. However the January 10th graph has an SE/Slope ratio greater than .5, which tells us that the trend reported could be due entirely to chance. The SE/Slope ratio for the average graph is lower than .5 showing statistically significant evidence for global warming.

PART B:



- How does this graph compare to the graphs from part A (i.e., in terms of the R² values, the fit of the resulting curves, and whether the graph

supports/contradicts our claim about global warming)? Interpret the results.

The newer graph has much lower SE/SLOPE and higher R^2 value which means we can draw stronger conclusions

from this graph and that the model explains a large portion of the variance.

The graph shows a clear increase in average temperature as the years increase.

The graph supports our claims about global warming.

- Why do you think this is the case?

The part B graph is produced by averaging out the temperatures across 21 cities.

This reduces local noise and grants more predictive power than sampling the temperature of one city.

- How would we expect the results to differ if we used 3 different cities?

What about 100 different cities?

Seeing as using 21 cities allowed us to reduce local noise, using 3 cities instead would lessen that effect.

The 3 cities model would be less predictive, having a higher SE/Slope and lower R^2 value, as less

noise is filtered out. Conversely, a 100 cities model would have even more random noise filtered out,

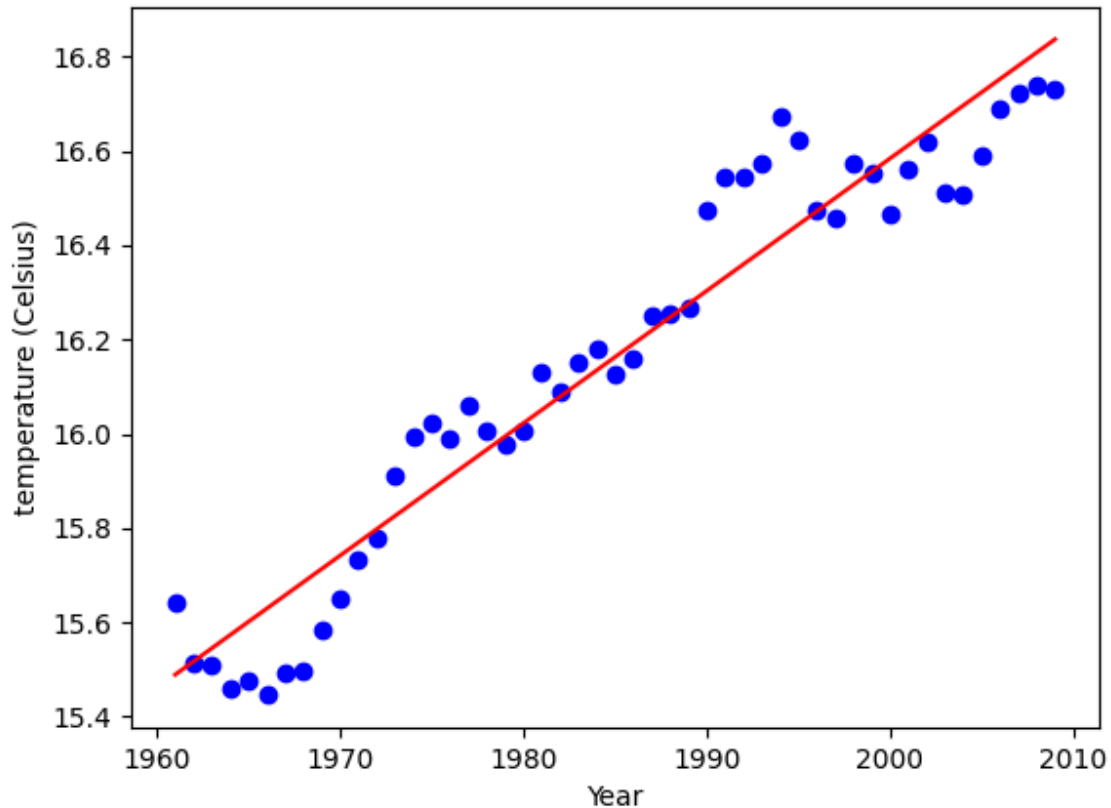
leading to a better and more predictive model.

- How would the results have changed if all 21 cities were in the same region of the United States (for ex., New England)?

If all 21 cities were in the same region, the resulting model would be hyper sensitive to random temperature fluctuations in that particular region resulting in worse predictive capacity for the rest of the United States.

PART C:

degree 1, R-square 0.924977562992991, SE/slope 0.041541393010224914



- How does this graph compare to the graphs from part A and B (i.e., in terms of the R² values, the fit of the resulting curves, and whether the graph supports/contradicts our claim about global warming)? Interpret the results.

The moving averages graph has an even higher R² value than any of the previous graphs, suggesting that its fit explains more of the variance compared to the previous plots.

It also has a much smaller SE/Slope value proving our fit's slope is statistically significant.

This plot shows temperature increases as the years increase, supporting our claim about global warming.

- Why do you think this is the case?

Our model uses moving averages of length 5 to average out temperature fluctuations sampled from 21 cities.

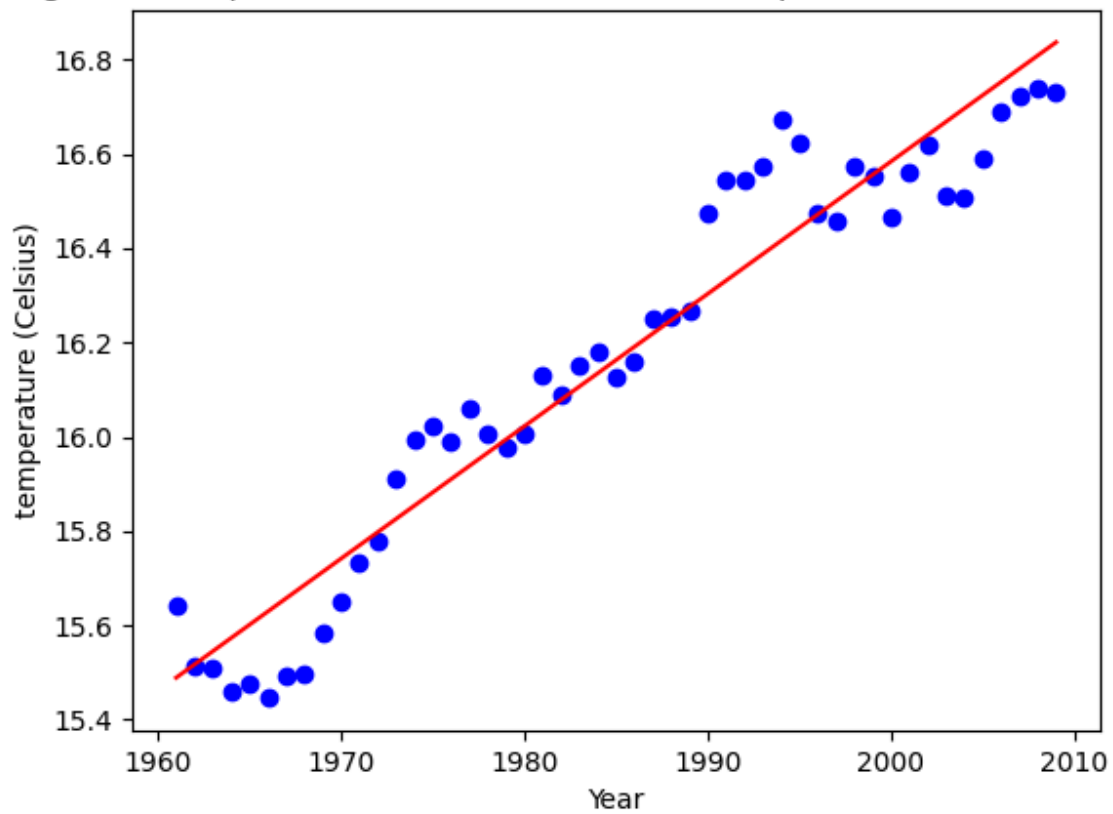
This removes noise and thus helps us better find the underlying trend. This technique removes

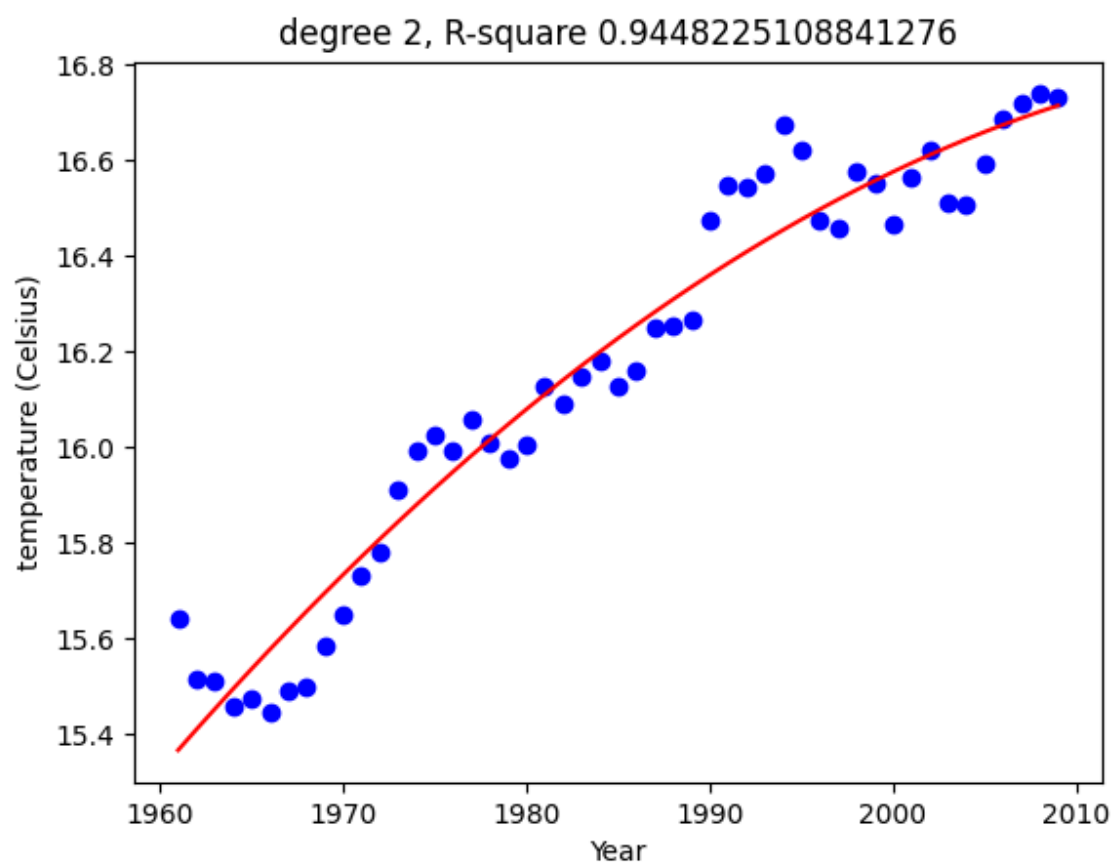
the most noise compared to sampling 21 cities (without moving averages) or sampling one.

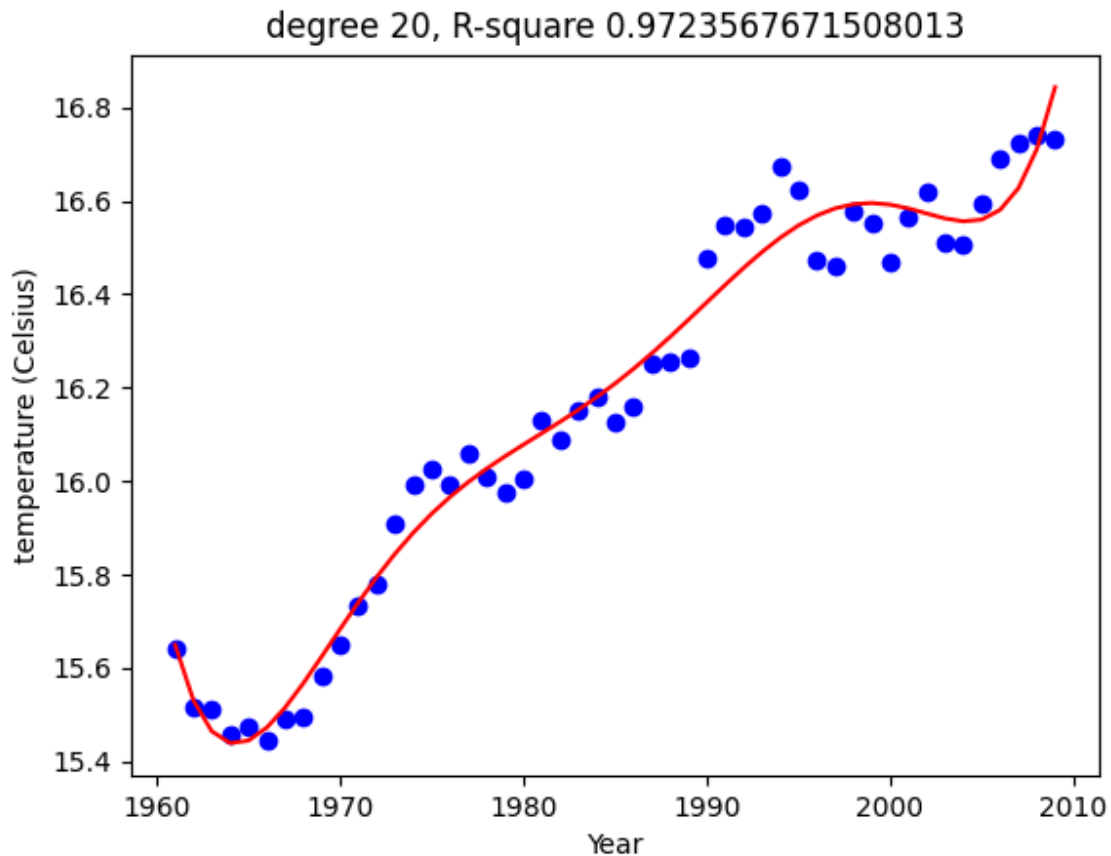
PART D:

PART I

degree 1, R-square 0.924977562992991, SE/slope 0.041541393010224914







- How do these models compare to each other?

The moving average graph of degree one as a straight line fit with an R^2 of .92.

The moving average graph of degree two has a slightly curvier line fit and an R^2 of .94.

The moving average graph of degree twenty has an extremely curvy line fit and an R^2 of .97.

- Which one has the best R^2 ? Why?

The moving average graph with degree 1 has the lowest R^2 value (.92), suggesting that its fit explains

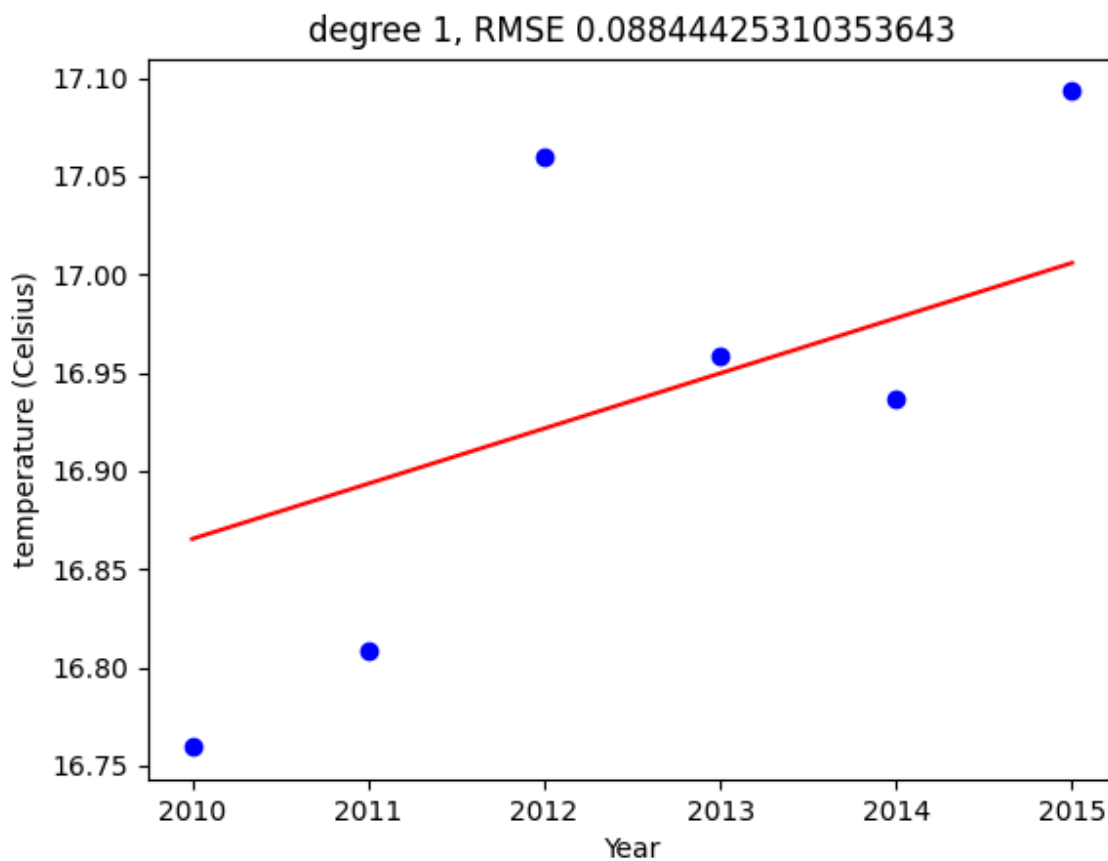
the least of the variance compared to the subsequent plots. As the degree increases to 2 and 20 the R^2

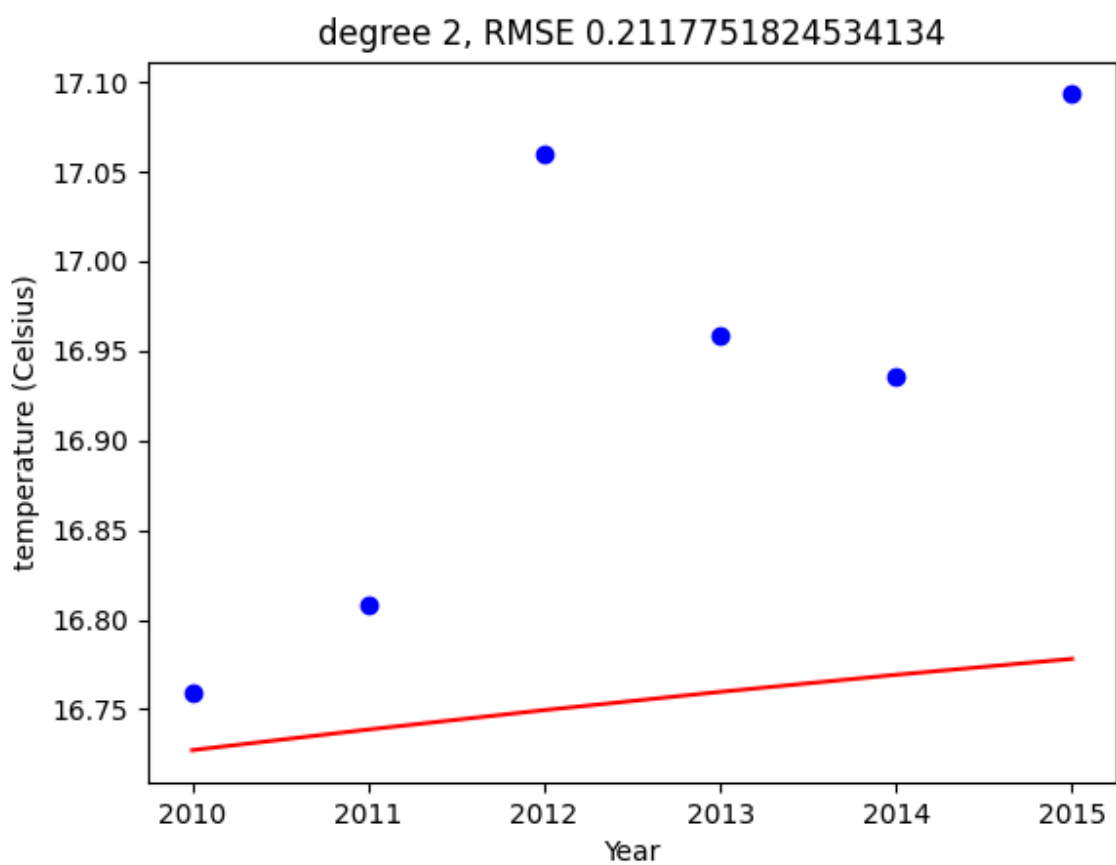
increases to .94 and .97 respectively. Proving that if we give the line of fit more parameters the model will have more flexibility to fit closely to the patterns of the data.

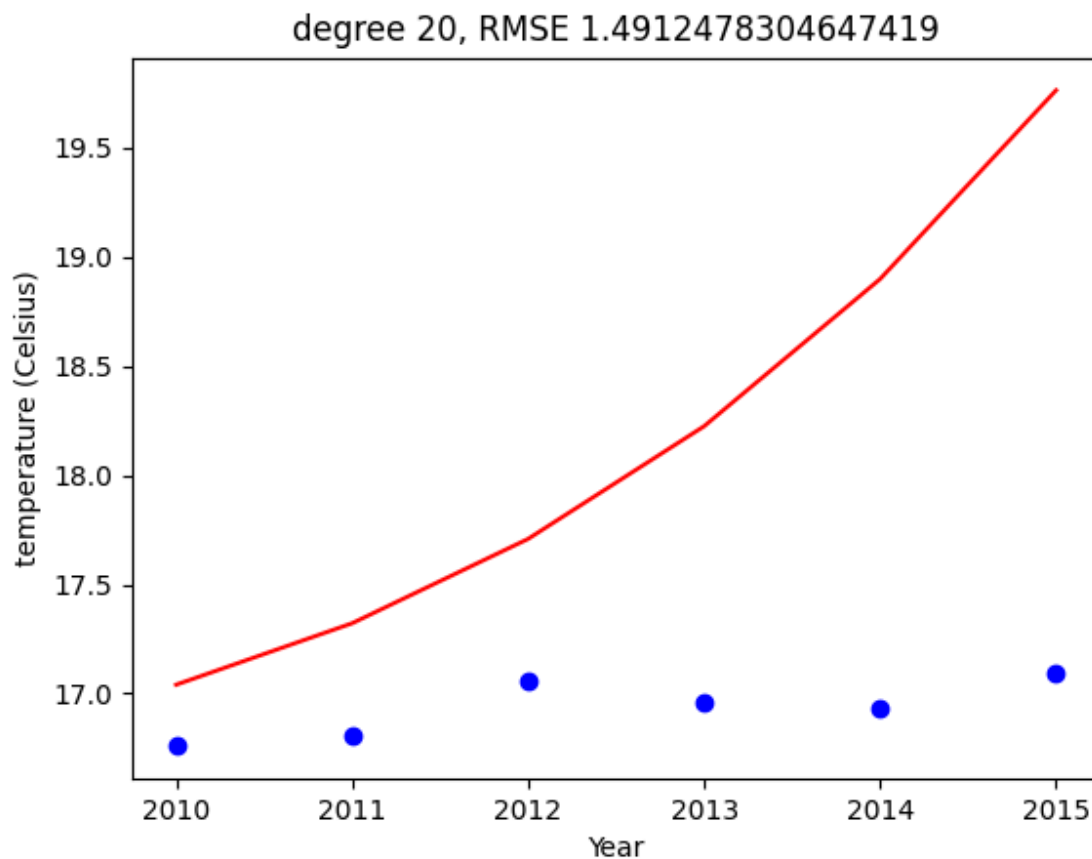
- Which model best fits the data? Why?

The model with degree 20 best fits the training data used to train the models. However, it's high R^2 is suspicious and hints that the model might've overfit to the data. We can't conclude which model generalizes best without seeing their performance on new data.

Part 2:







- How did the different models perform? How did their RMSEs compare?

The models with degree 1 performed the best with an RMSE of .08. The models with a degree of 2 performed second best with an RMSE of .211. The models with degree 20 performed the worst with an RMSE of 1.49.

- Which model performed the best? Which model performed the worst? Are they the same as those in part D.2.I? Why?

The model with degree 1 performed the best in D.2.II whilst the model with degree 20 performed the best on D.2.I. This is a clear case of overfitting, where our more complex and flexible models fail to generalize to new data. The models that best fit the training data do not necessarily always perform the best on the test data.

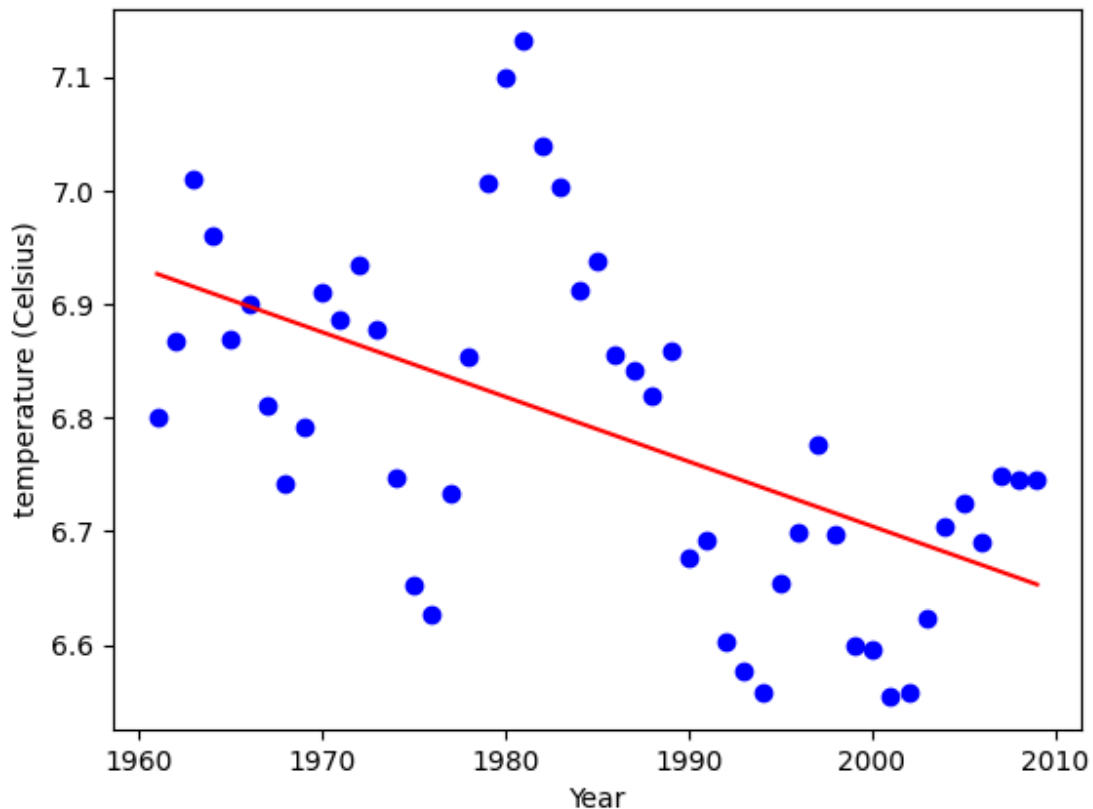
- If we had generated the models using the A.4.II data (i.e. average annual temperature of New York City) instead of the 5-year moving average over 22 cities, how would the prediction results 2010-2015 have changed?

Since we'd no longer use moving average and rely on the temperature of one city (instead of various ones),

the results of our models would be less resistant to stochastic variation and noise. The resulting models would perform worse on new data.

PART E:

degree 1, R-square 0.3003867469373056, SE/slope -0.22260745326727094



- Does the result match our claim (i.e., temperature variation is getting larger over these years)?

Our results contradicts the claim, as the slope of the regression line is negative. Which means that the standard deviation of the temperatures are decreasing over time, so temperature variation is going down.

- Can you think of ways to improve our analysis?

We can increase the number of cities that we're using to give a more representative picture of the national trend. We can increase the size of the moving average window to smooth out and decrease more noise.